

DOCUMENTATION TECHNIQUE

Documentation Technique & Architecture du Pipeline

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Football Match Manager - Technical Specification & Architecture Document

1. Executive Summary & Architecture Overview

1.1 Executive Brief

The Football Match Manager is a RESTful API designed to manage football matches, leagues, and teams, featuring a user-following mechanism for match tracking. The system utilizes a JWT-based authentication pattern to protect user-specific data and implements strict rate-limiting to ensure service stability. It serves as a structured data gateway for sports-related entities and user interactions.

1.2 Maturity Assessment

The specifications are technically robust regarding the API contract, evidenced by a high health index and complete functional mapping of endpoints. While the project is **READY** for execution, a minor **REFINEMENT** is suggested to explicitly define the system boundaries (Scope & Out-of-Scope) and resolve low-priority structural gaps to prevent scope creep during development.

1.3 Technical Stack

- **API Architecture:** RESTful API
- **API Versioning:** v1
- **Authentication:** JWT (JSON Web Token)
- **Data Format:** JSON
- **Identifier Standard:** UUID

1.4 Architectural Constraints

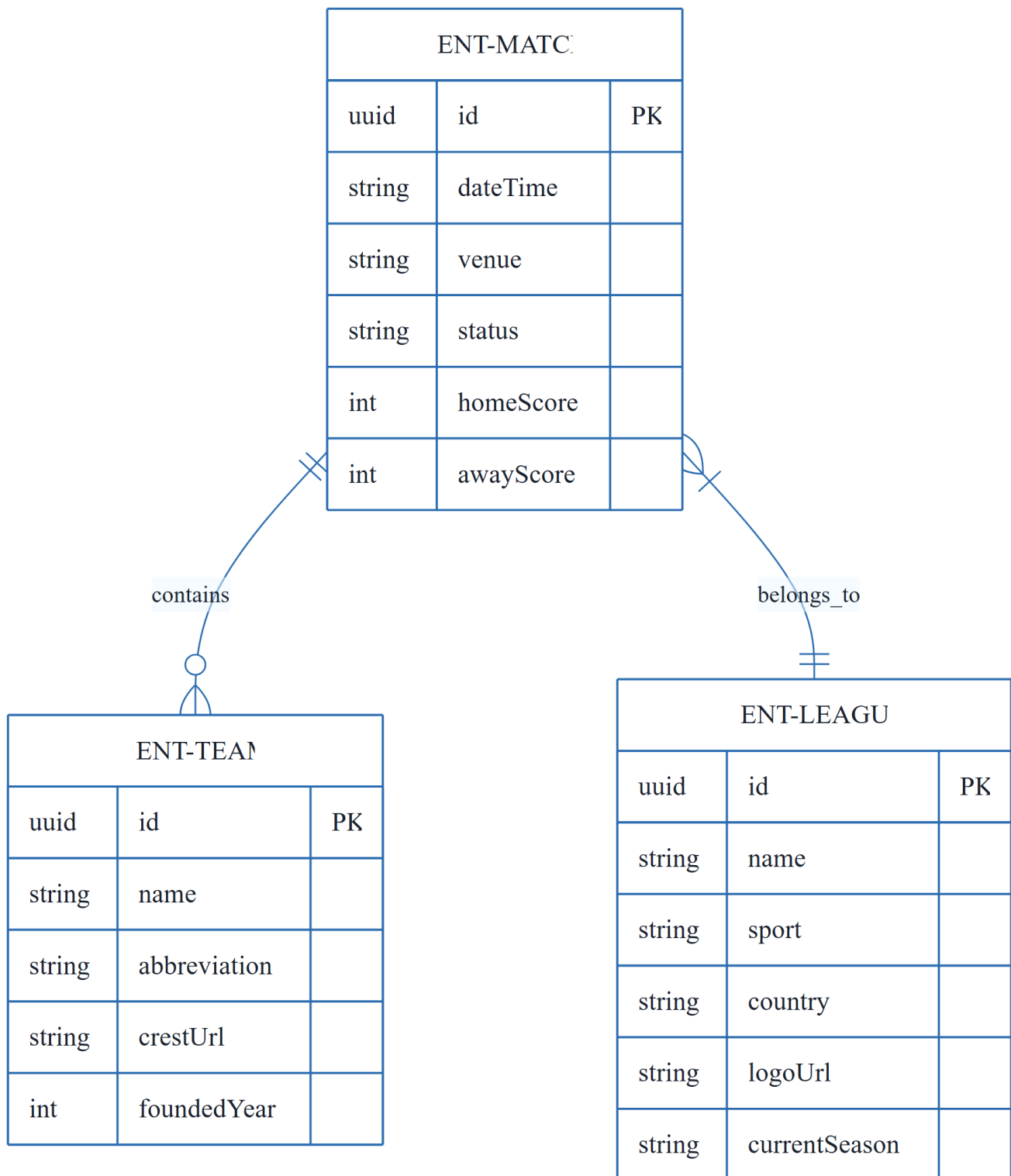
- **API Versioning:** Hard-coded path prefix `/api/v1/`.
- **Authentication:** Bearer JWT token required for all non-auth endpoints.
- **Rate Limit (Auth):** Maximum 5 requests per minute per IP.
- **Rate Limit (General):** Maximum 60 requests per minute per user.
- **Pagination:** Default page 1, default limit 20 for match lists; default limit 50 for team searches.
- **Data Validation:** `matchId` must be a UUID; `dateFrom` and `dateTo` must be ISO 8601 strings.

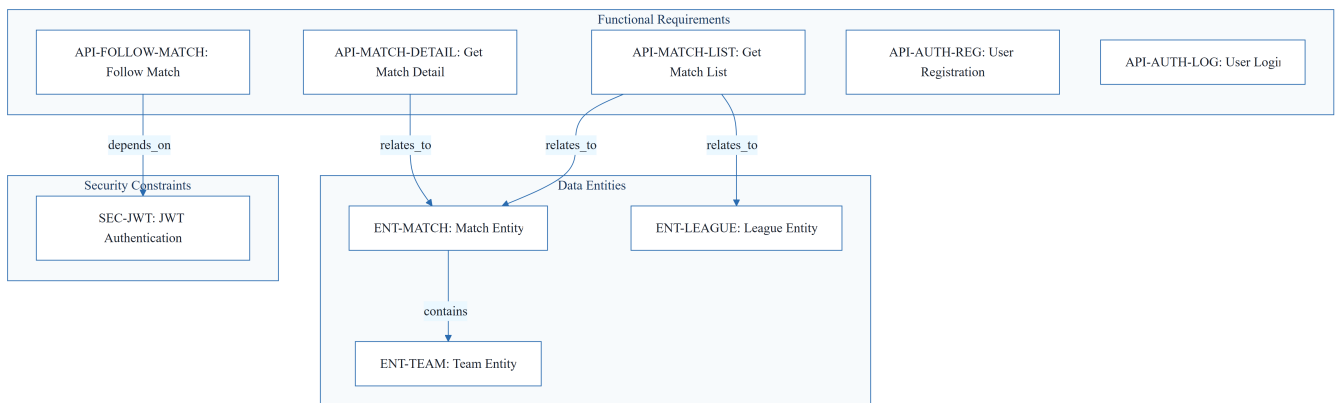
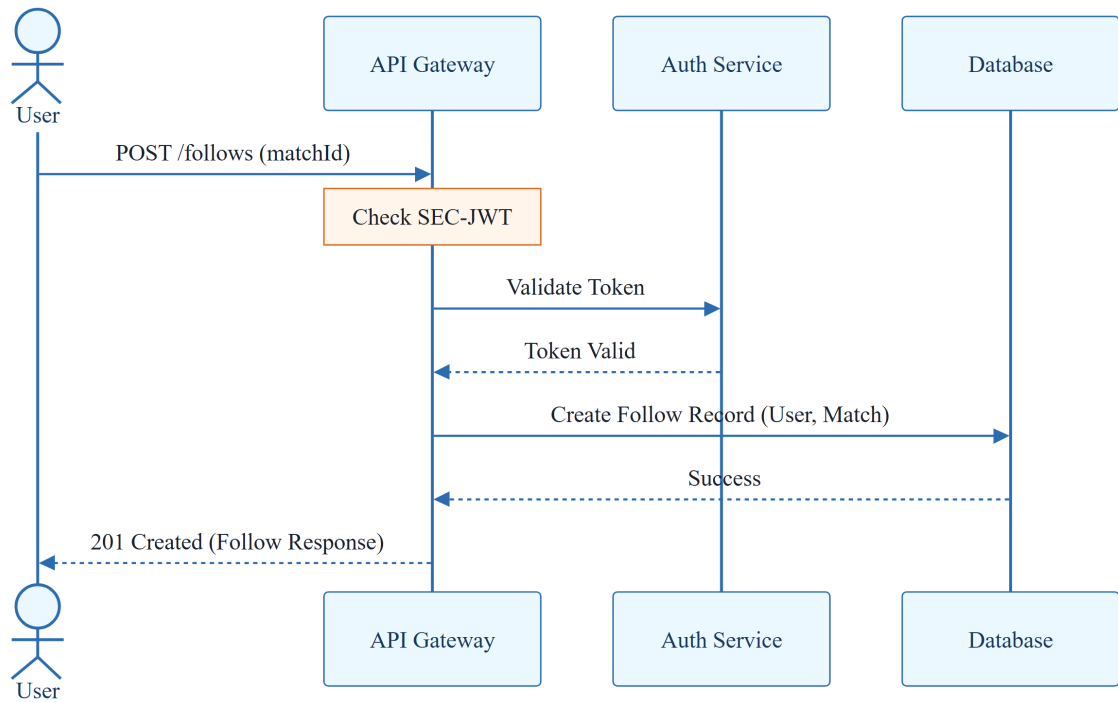
1.5 Critical Dependencies

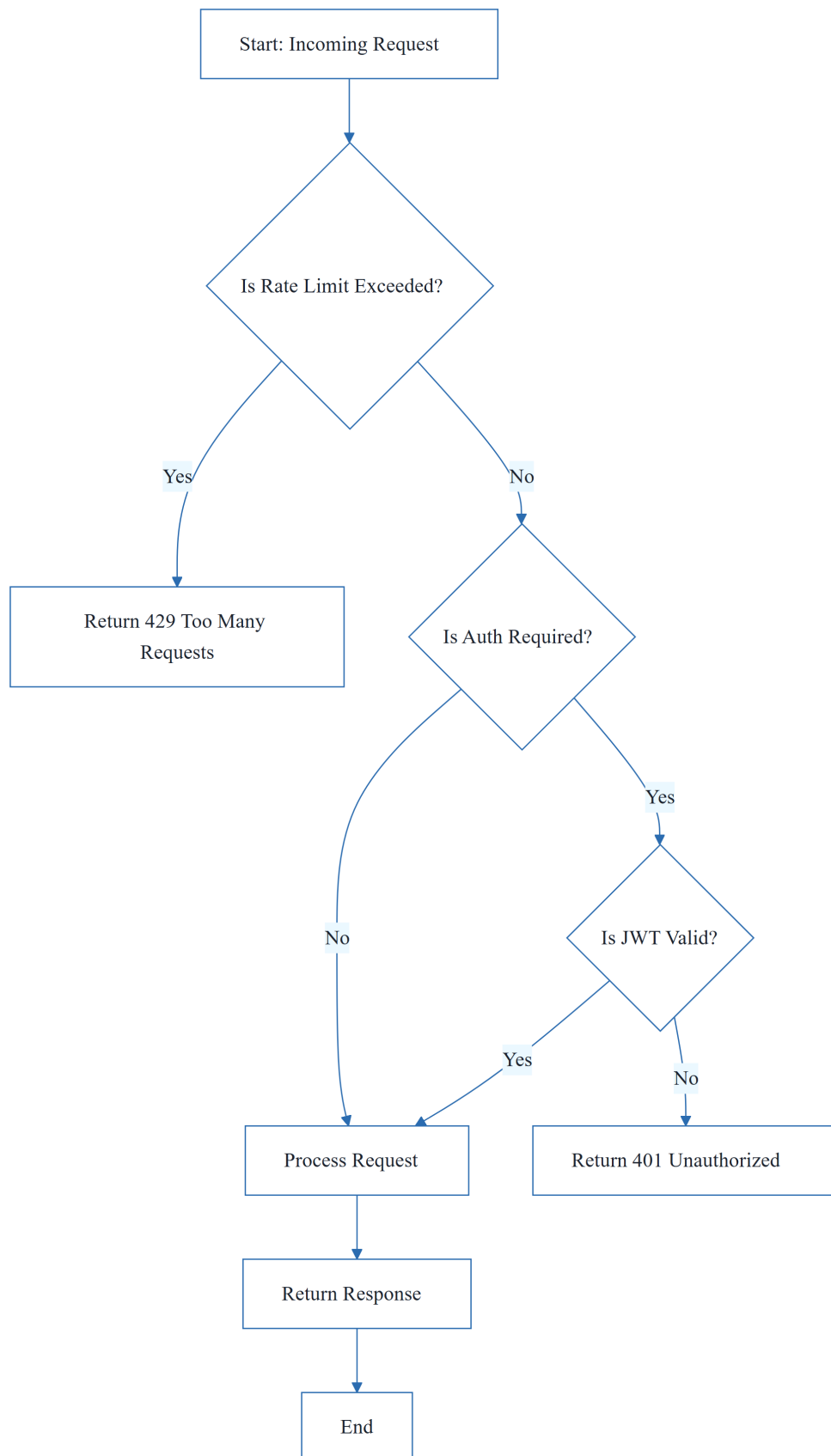
- **JWT (JSON Web Token):** Essential for session management and endpoint authorization.
- **Entity Relationships:** Match entity depends on Team and League entities for data integrity.

- **User Identity:** Follows mechanism depends on User identity and Match UUID.
- **Network Tracking:** IP-based tracking required for authentication rate-limiting gates.

2. Architecture Workflows

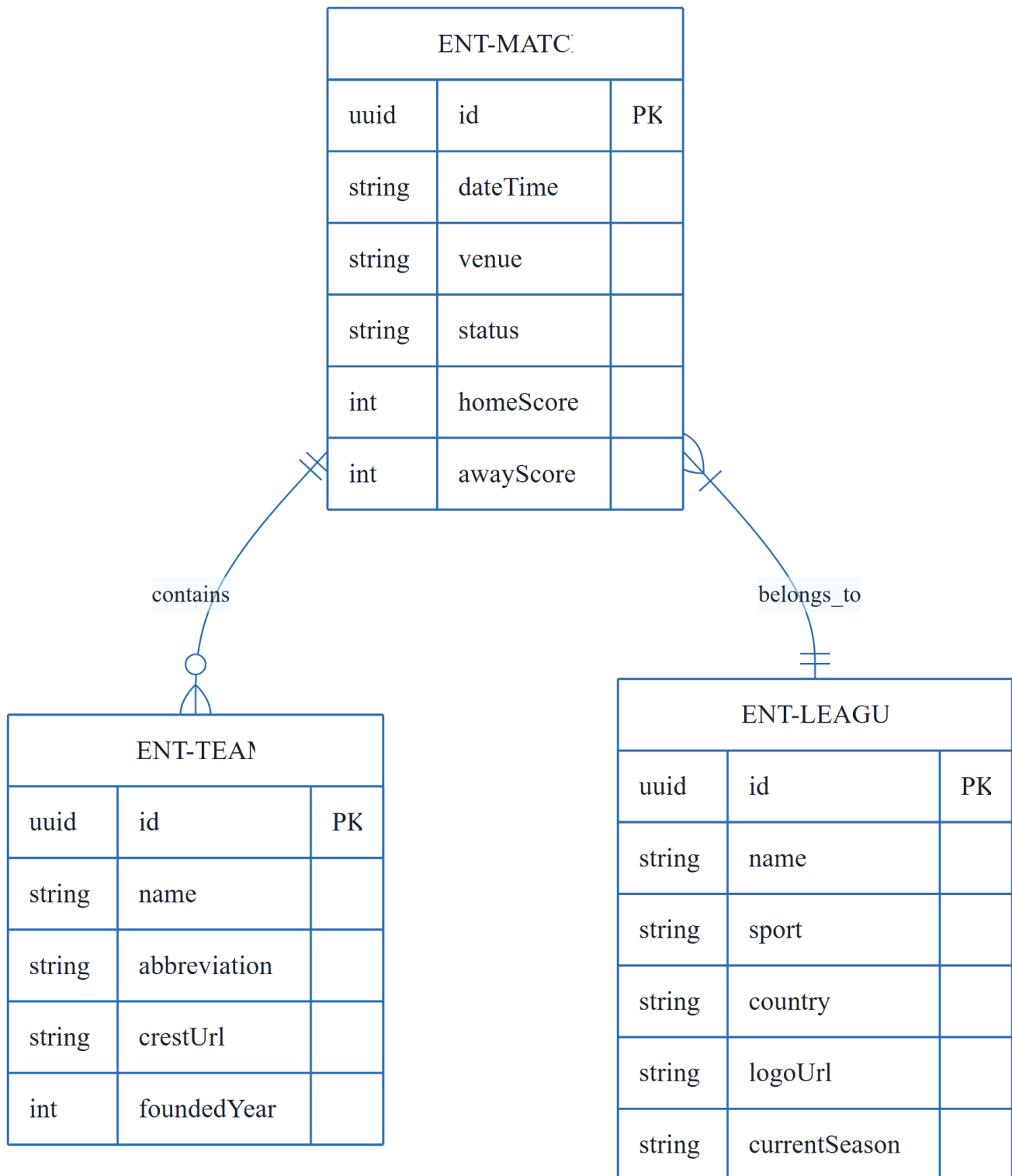




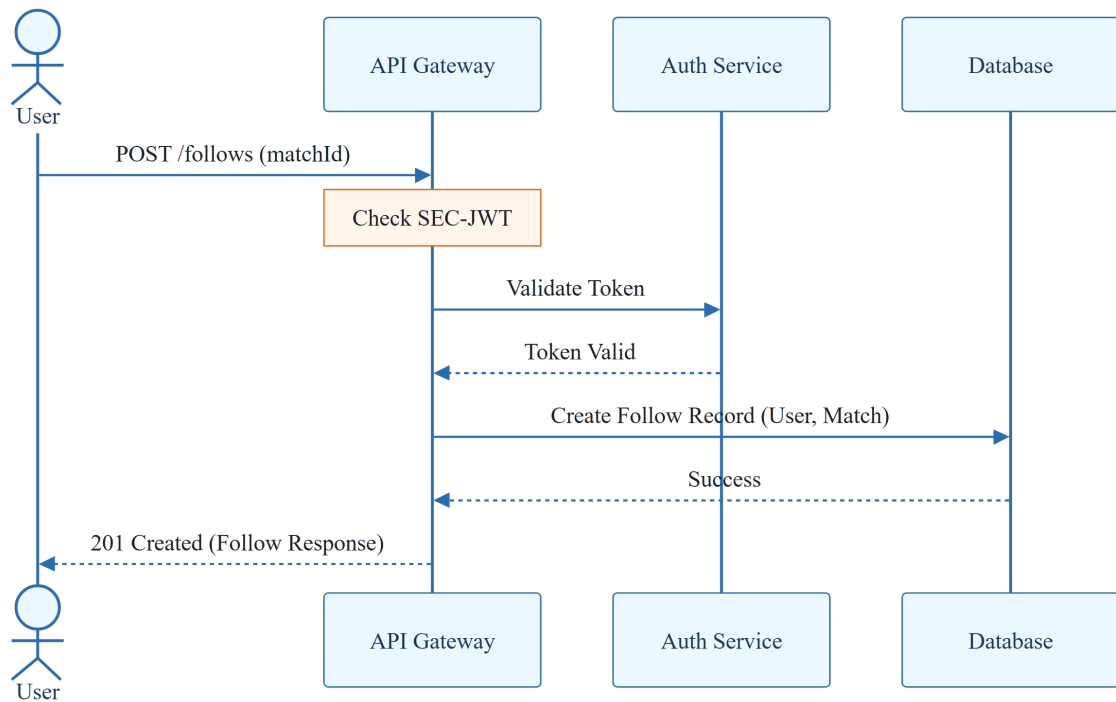


& Visual Diagrams

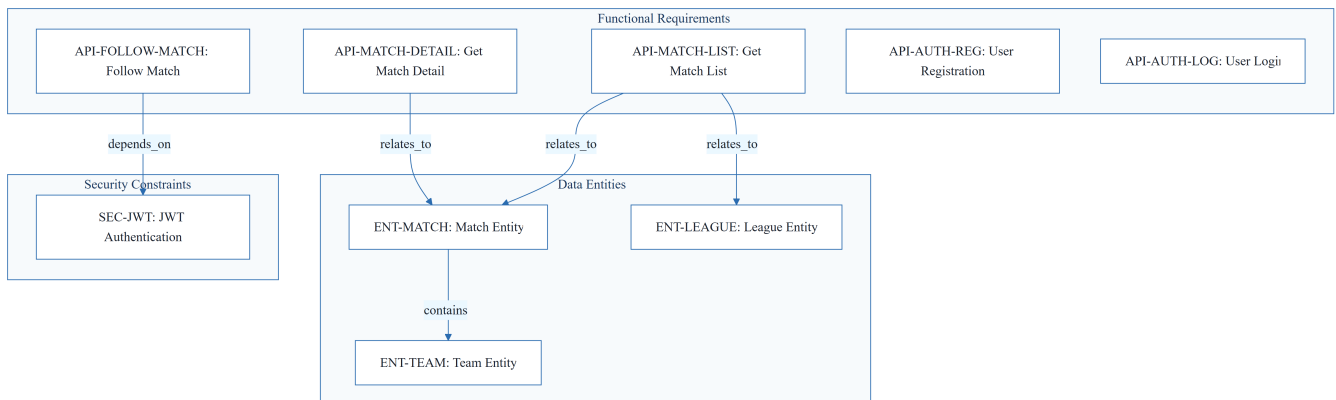
2.1 Data Model (ER Diagram)



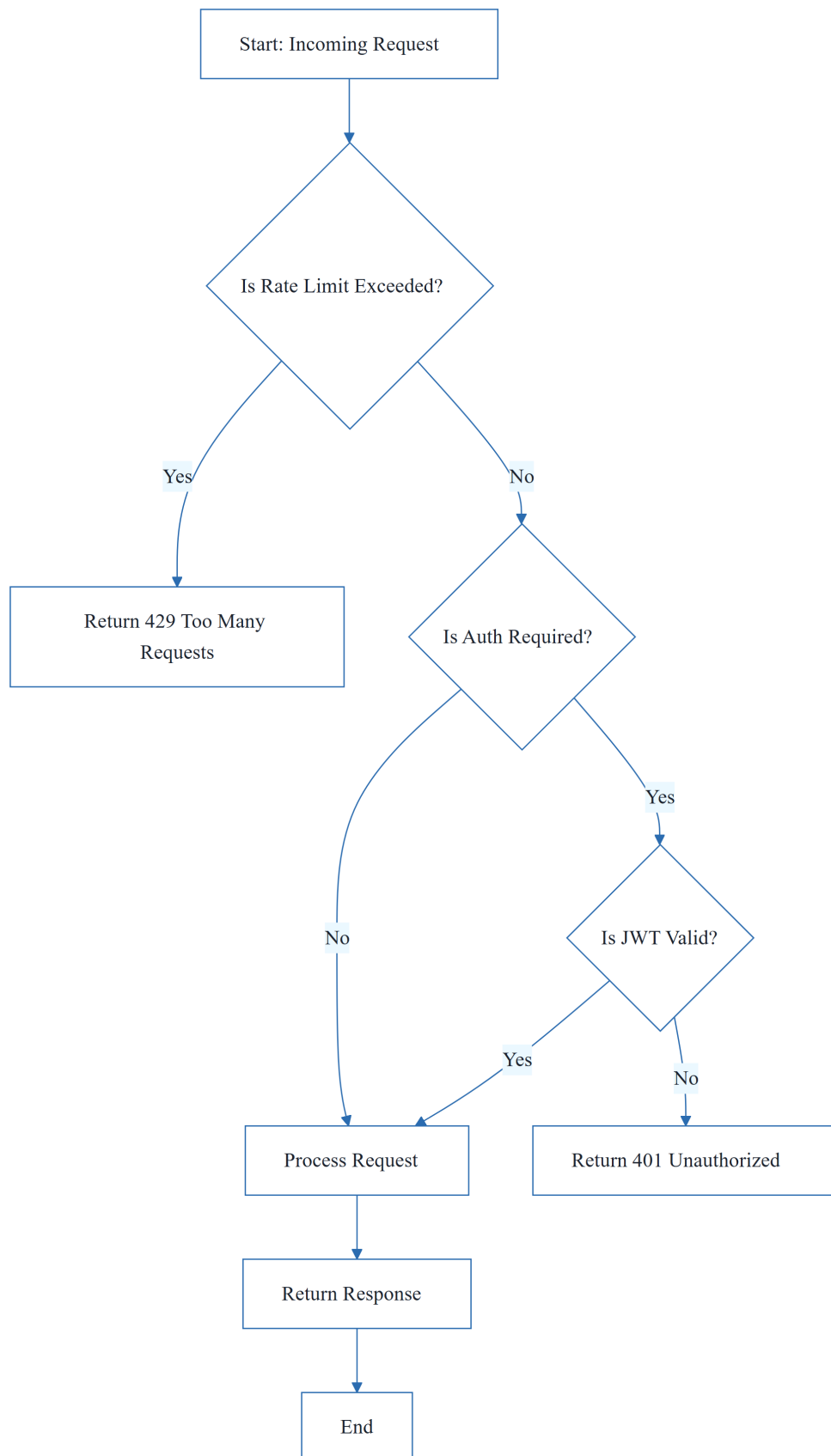
2.2 Match Following Sequence



2.3 Requirements Traceability Map



2.4 API Request Processing Workflow



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

ID	Requirement Type	Description	Source Section
API-AUTH-REG	Functional	Allow user registration via POST /auth/register	Authentication
API-AUTH-LOG	Functional	Allow user login to receive JWT token via POST /auth/login	Authentication
API-MATCH-LIST	Functional	Retrieve a list of matches with filtering by leagueId, date, status, and team name via GET /matches	Matches
API-MATCH-DETAIL	Functional	Retrieve a specific match by its unique ID via GET /matches/:id	Matches
API-FOLLOW-MATCH	Functional	Allow a user to follow a match via POST /follows	User Follows
API-UNFOLLOW-MATCH	Functional	Allow a user to unfollow a match via DELETE /follows/:matchId	User Follows
SEC-JWT	Non-Functional	All endpoints (except auth) require a valid JWT token in the Authorization header.	Authentication
NFR-RATE-LIMIT	Non-Functional	Auth endpoints limited to 5 req/min/IP; others 60 req/min/user.	Rate Limiting

3.2 Security Rules

- **Authentication Mechanism:** Bearer Token (JWT).
- **Authorization Scope:** All endpoints except /auth/* require a valid token.
- **Rate Limiting:**
 - Authentication endpoints: 5 requests per minute per IP.
 - General endpoints: 60 requests per minute per authenticated user.

3.3 Data Models

ENT-MATCH (Match)

```
{
  "id": "uuid",
  "homeTeam": { "..."},
  "awayTeam": { "..."},
  "dateTime": "ISO string",
  "venue": "string",
  "league": { "..."},
  "status": "upcoming|live|finished|postponed|cancelled",
  "homeScore": "integer (nullable)",
  "awayScore": "integer (nullable)",
  "lastUpdated": "ISO string"
}
```

ENT-TEAM (Team)

```
{
  "id": "uuid",
  "name": "string",
  "abbreviation": "string",
  "crestUrl": "url",
  "foundedYear": "integer"
}
```

ENT-LEAGUE (League)

```
{
  "id": "uuid",
  "name": "string",
  "sport": "string",
  "country": "string",
  "logoUrl": "url",
  "currentSeason": "string"
}
```

User (Authenticated)

```
{
  "id": "uuid",
  "email": "string",
  "displayName": "string",
  "avatarUrl": "url"
}
```

Follow Response

```
{
  "userId": "uuid",
  "matchId": "uuid",
  "followedAt": "ISO string",
  "notificationsEnabled": "boolean"
}
```

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Missing Section	Priority	Remediation Advice
Scope & Out-of-Scope	MEDIUM	Define what the API will NOT handle (e.g., payment for tickets, real-time player stats).
Open Questions & Uncertainties	LOW	Identify any undecided endpoints or data model fields.

4.2 Remediation & Workflow

To ensure the project remains within scope and avoids creep, the development team must first document the "Out-of-Scope" boundaries. Following this, a final review of the data models should be conducted to identify any missing fields before the implementation phase begins.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
API	TECHNICAL_STACK	API Contract: Football Match Manager	The set of RESTful endpoints versioned as v1 for managing sports data and user interactions.
CORS Standard	TECHNICAL_STACK	Base URL	The cross-origin resource sharing policy implicitly required to allow browser-based clients to interact with the specified backend endpoints.
ID	BUSINESS_DOMAIN	API-MATCH-DETAIL	The unique identifier used to retrieve a specific sports event record.

Term	Category	Context Anchor	Project Definition
IP	TECHNICAL_STACK	NFR-RATE-LIMIT	The network address used as a key to apply a maximum of 5 requests per minute for authentication endpoints.
JSON	TECHNICAL_STACK	Data Models	The lightweight data-interchange format used for all request bodies and response payloads.
JWT	TECHNICAL_STACK	SEC-JWT	The signed token passed in the Authorization header as a Bearer credential to secure non-auth endpoints.
UUID	TECHNICAL_STACK	ENT-MATCH	The 128-bit universally unique format used for primary keys across all entities including matches, teams, and leagues.
dateFrom	BUSINESS_DOMAIN	API-MATCH-LIST	The ISO string parameter serving as the lower temporal boundary for filtering sports events.
dateTo	BUSINESS_DOMAIN	API-MATCH-LIST	The ISO string parameter serving as the upper temporal boundary for filtering sports events.
leagueId	BUSINESS_DOMAIN	API-MATCH-LIST	The unique reference to a specific sports competition used to filter the list of events.